

Student Enrollment B-Tree Scenario

A university records system indexes student IDs into a B-tree of order $t=2$. The tree is initially loaded with the following student IDs: 15, 25, and 35.

During the enrollment period, the following actions occur:

1. We insert student ID 10.
2. We insert student ID 20.
3. We insert student ID 5.
4. We insert student ID 30.
5. We insert student ID 40.

After the initial insertions, we perform a search for student ID 20 to verify their enrollment.

Then we search for student ID 99 to confirm they are not enrolled.

Student ID 25 drops out, so we delete them from the system.

Student ID 5 transfers to another university, so we delete them.

Finally, a late enrollment occurs, and we insert student ID 50.